Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/22/2025
PI Name:	Lee, Peter M.D., Ph.D.		
Project Title:	Development of a Novel Tissue Engineered Skeletal Muscle Model of Atrophy Using Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	43210-1240	**Congressional District:**	3
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/25/2020 (original end date was 10/25/2019) per NSSC information (Ed., 7/28/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0137		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	It is well known that extended spaceflight causes significant skeletal muscle atrophy in astronauts. However, the mechanisms that drive this process are not well understood. Previous space shuttle experiments have shown that spaceflight leads to atrophy in isolated tissue engineered skeletal muscle. However, we do not know how muscle tissues sense the microgravity environment leading to the muscle atrophy. Since these earlier flights with tissue engineered skeletal muscle, we have developed the ability to miniaturize these functional muscle units and to also measure real-time forces. Advances in proteomics and genomics have also significantly advanced our ability to better understand molecular mechanisms of muscle pathology. We propose ground-based studies to further develop these miniaturized tissue-engineered skeletal muscle constructs as an ideal three-dimensional, functional, physiologically relevant, in vitro model for studying microgravity-induced skeletal muscle atrophy. We will validate a previously developed atrophy model using larger muscle constructs based on tension reduction by investigating the potential use of a 3D clinostat as a valid ground-based model for simulating the effects of microgravity. These data will provide the basis for potential future spaceflight studies that will allow investigators to measure real-time functional data in muscle as they atrophy. It will also enable the evaluation of various atrophy countermeasures during spaceflight. Our hypothesis is that simulated microgravity induces atrophy in tissue engineered skeletal muscle and can be attenuated by growth factors and pharmaceuticals. The successful completion of this proposal will establish the development of an efficient, high throughput, ground-based, in vitro model system for studying the effects of microgravity on skeletal muscle.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Per NASA Kennedy Space Center (KSC), this project has closed. No final reporting received. Please see Bibliography for publications resulting from this funding (Ed., 8/2/21).
Bibliography Type:	Description: (Last Updated: 07/22/2025)
Articles in Peer-reviewed Journals	Ren Z, Ahn EH, Do M, Mair DB, Monemianesfahani A, Lee PHU, Kim DH. "Simulated microgravity attenuates myogenesis and contractile function of 3D engineered skeletal muscle tissues." npj Microgravity. 2024 Feb 16;10(1):18. https://doi.org/10.1038/s41526-024-00353-z ; PMID: 38365862; PMCID: PMC10873406 , Feb-2024